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The palaeolatitude controversy in the Silurian of Newfoundland resolved: new palaeomagnetic results from the central mobile belt

M.A. Smethurst*, S.A. McEnroe

Geological Survey of Norway, N-7491 Trondheim, Norway

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Abstract

A detailed palaeomagnetic investigation of early Silurian volcanics of the Lawrenceton Formation in the Central Mobile Belt (CMB) of Newfoundland reveals primary remanence that places the sampled part of the CMB on the margin of Laurentia in low southerly latitudes in the early Silurian. The palaeomagnetic data indicate large vertical axis (net clockwise) rotations within the CMB and between the CMB and cratonic Laurentia. The volcanics were folded, faulted and in places cleaved in Silurian to Early Devonian times. Anisotropy of magnetic susceptibility measurements on lava flow samples verify field observations of a primary flow-parallel foliation and confirm field measurements of the palaeo-horizontal. A well-developed tectonic fabric was detected at one sampling site near the Reach Fault.

Silurian redbeds of the Big Indian Pond Formation were also sampled for palaeomagnetic study. The sediments are folded, faulted and cleaved. A statistically sound syn-folding remanence was identified which, in a tectonically corrected state, is consistent in direction with the remanence in the Silurian volcanics. The sampled sediments are more intensely deformed than the volcanics and magnetic anisotropy measurements indicate both primary layering- and secondary cleavage-related fabrics in varying degrees from site to site. It is possible that cleavage development resulted in remanence reorientation leading to the illusion of syn-folding remanence when, in fact, the remanence was acquired prior to folding.

Previously published palaeomagnetic investigations of Silurian rocks in the CMB revealed a discrepancy in palaeofield inclination between volcanic and sedimentary rock types. The sedimentary rocks yielded shallow inclinations and the volcanics intermediate inclinations. Inclination shallowing in the sediments was put forward as an explanation, as well as contamination of the sedimentary results with a shallow overprint. Others argued that the results from the volcanics were in error. The present investigation demonstrates that key volcanics carry a shallower remanence than previously thought, bringing them into approximate agreement with the majority of the results from sediments. Steep results from the Springdale volcanics and very shallow results from the King George IV lake sediments are at variance with the rest of the data. The Springdale result has large error limits that overlap with the other data. The anomalous King George IV result remains an outlier.
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1. Introduction

In the late 1980s and 1990s, palaeomagnetic investigation focused on rocks in the Central Mobile Belt of

* Corresponding author.
E-mail address: mark.smethurst@ngu.no (M.A. Smethurst).

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Newfoundland (van der Pluijm and van Staal, 1988), with a view to charting the closure of the early Palaeozoic ocean tract between the Avalon terrane and Laurentia (Fig. 1). It was expected that palaeomagnetic investigation of subaerial volcanic rocks and sediments of early Silurian age would reveal that the CMB was close to the Appalachian margin of Laurentia and the Avalon terrane at that time and, therefore, that this part of the Iapetus Ocean was essentially closed in the early–mid-Silurian.

The palaeomagnetic investigations unveiled a significant discordance between palaeomagnetic field inclinations recorded in early Silurian volcanic rocks and coeval red sediments of the CMB. The volcanic rocks of the Botwood Group (Lawrenceton Formation) (Lapointe, 1979; Gales et al., 1989) and the Springdale Group (Potts et al., 1993) yielded intermediate palaeomagnetic field inclinations that put the CMB in moderate southerly latitudes in the early Silurian (~ 25°S). The shallow palaeofield inclination carried by the Botwood Group sediments (Wigwam Formation) (Lapointe, 1979; Buchan and Hodych, 1992), Springdale group sediments (Potts et al., 1993; Hodych and Buchan, 1994a) and King George IV Lake sediments (Buchan and Hodych, 1989) place the CMB near the early Silurian equator (~ 10°S). A lively debate ensued in the literature between followers of the results from the volcanic rocks (van der Pluijm et al., 1993; Stamatakos et al., 1994, 1995) and advocates of the sedimentary results (Buchan and Hodych, 1993; Hodych and Buchan, 1994b).

Upon reporting modest southerly palaeolatitudes for the volcanic rocks, Van der Pluijm et al. (1990), Gales et al. (1989) and Potts et al. (1993) all noted their consistency with reference data from Laurentia and consequently proposed a Silurian position for the CMB on the Appalachian margin of Laurentia. This interpretation was most recently restated by Stamatakos et al. (1995) who, adopting the APW path for Laurentia of Van der Voo (1993), pointed out that a moderate southerly latitude for the CMB is consistent with a position on the Appalachian margin of Laurentia, whereas an equatorial position for the CMB dictated by the sediments would require a very considerable longitudinal offset of the CMB to avoid significant overlap with Laurentia. Buchan and Hodych (1992, 1993) challenged the integrity of the reference data used by the aforementioned authors and proposed an equatorial position for the margin of Laurentia and the CMB in the early Silurian in accordance with the results from the sediments and some lava flows of Newfoundland (Buchan and Hodych, 1992, 1993).

The essential factors at the centre of the controversy are twofold: the reliability of the remanent magnetisations carried by the Silurian rocks of the CMB, and the reliability and definition of the APW path for Laurentia. Van der Pluijm et al. (1993) argued that the palaeomagnetic results from the sediments are affected by compaction and related inclination shallowing (for inclination shallowing, see, e.g. Deamer and Kodama, 1990; Sun and Kodama, 1992). In response, Buchan and Hodych (1993) and Hodych and Buchan (1994a,b) argued that inclination shallowing in the sediments is not significant and that the results from the volcanic rocks are based on samples from few sites in which tectonic corrections are in doubt. Stamatakos et al. (1995) reexamined some of the sedimentary rocks and noticed that the small collection of fine-grained sediments carried a steeper remanence than the coarse sediments, almost as steep as the results from the volcanic rocks, and concluded that these steeper directions are a true record of the early Silurian palaeofield in the sediments. The observations of Stamatakos et al. (1995) undermine compaction and inclination shallowing as a viable explanation for the overall shallow remanence in the sediments. Instead, Stamatakos et al. suggested that the shallow directions in the coarser sediments might be due to incomplete isolation of remanence components and/or shallowing related to acquisition of a DRM (for shallowing of DRM, see, e.g. King, 1955). The investigation of Stamatakos et al. left the controversy apparently solved in favour of the steeper results.

This situation was short-lived because at around the same time, MacNiocaill and Smethurst (1994) published an APW path for Laurentia that differed significantly from that of Van der Voo (1993). The new path, later refined by Torsvik et al. (1996), suggested an equatorial position for the Laurentian margin in the early Silurian. In this case, the shallow results from the sediments put the CMB in its most likely position close to the margin of Laurentia and the steeper results from the volcanic rocks imply an unlikely latitudinal (oceanic) separation between the CMB and Laurentia in the early/mid-Silurian. Adoption of the revised path essentially inverts the palaeomagnetic discrepancy, putting

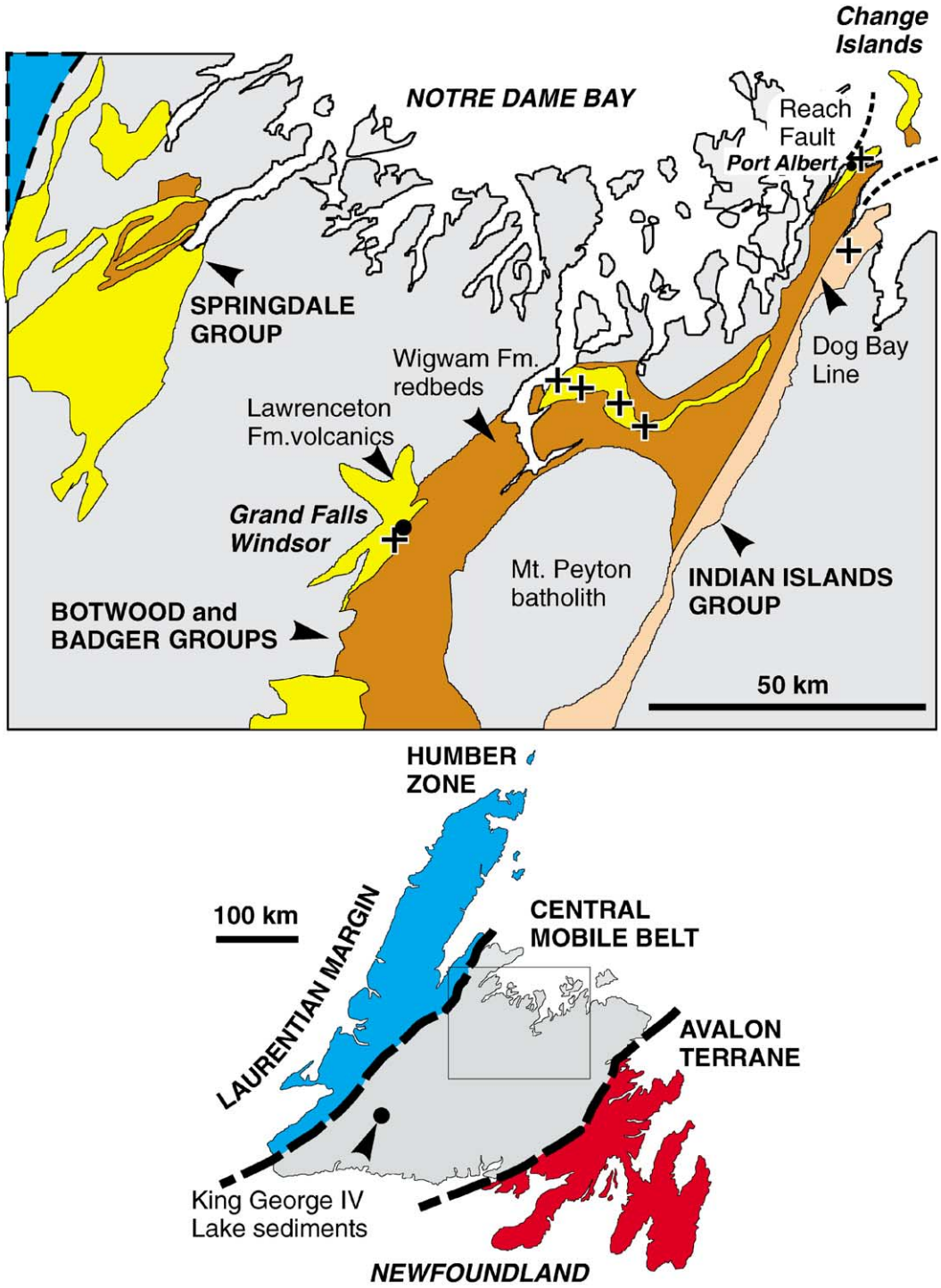


Fig. 1. Simplified geological maps of Newfoundland and the Notre Dame Bay region. Yellow: Silurian volcanics. Brown: Silurian and Late Ordovician sediments. Crosses: sampling regions of this study.

the results from the sediments in line with geological evidence for contiguity between the CMB and Laurentia by late Silurian time (van der Pluijm et al., 1990; Williams et al., 1993), leaving the steeper results from the volcanic rocks as anomalies.

The history outlined above shows that in 1996 the Silurian controversy was very much alive and further investigation was required to solve it. A great deal of attention had been paid to the palaeomagnetic signature of the sediments and inclination shallowing, leaving the volcanic rocks to be represented by a mixture of old and new studies comprising results from comparatively few and widely separated sampling sites. Also disturbing was the apparent absence of reversals in the Botwood volcanics (all reversed) and Springdale volcanic rocks (all normal). Consequently, Smethurst chose to carry out a new and extensive study of the volcanic rocks in the Lawrenceton Formation of the Botwood group. He also realised that there were early Silurian red sediments in northeast Newfoundland that previously had not been studied palaeomagnetically. They belong to the Big Indian Pond Formation (Centennial Formation) of the Indian Islands Group (Williams, 1972) located east of the Dog Bay Line (Williams, 1993; Williams et al., 1993). These rocks were also sampled for palaeomagnetic analysis.

Once again, the controversy appears to be solved. We show that the volcanic rocks were undersampled and that both the sediments and volcanic rocks yield shallow remanence directions. Data from both rock types agree and imply a low-latitude position on the Appalachian margin of Laurentia for all sampled parts of the CMB. Laurentia is positioned according to a minor revision of the APW path of Torsvik et al. (1996). We hope that the present work will prove to be an enduring solution to the drawn out palaeomagnetic controversy surrounding the Silurian rocks of Newfoundland.

2. Geological setting and sampling

2.1. Volcanic rocks: Lawrenceton Formation, Botwood Group

2.1.1. Rock description

The Lawrenceton Formation (Figs. 1 and 2) consists of subareal rhyolitic lava flows, tuffaceous sandstones and conglomerates (Dean, 1978). The lava flows are maroon in colour, rich in hematite and commonly contain feldspar phenocrysts and calcite-filled vesicles. Lava flow tops are commonly brick-red in colour due to extensive oxidation during lava flow emplacement or subsequent weathering. Results of postemplacement breakdown of primary iron silicates and oxides to specular hematite and hematite pigment are observed throughout the lava flows. Subaerial sediments of the Wigwam Formation conformably overlie the Lawrenceton Formation.

2.1.2. Age

Fossils and radiometric ages constrain accumulation of the Lawrenceton and Wigwam Formations to Llandovery–Wenlock (e.g. Williams, 1972; Arnott et al., 1985; van der Pluijm et al., 1987; Williams et al., 1993).

2.1.3. Deformation

The Silurian rocks of northeast Newfoundland were folded about northeast-trending and horizontal axes in mid- to late Silurian times (Karlstrom et al., 1982; Williams et al., 1993) and underwent brittle thrusting and faulting in latest Silurian to early Devonian (Williams et al., 1993). The folded Botwood Group is cut in the northwest by the Reach Fault, and in the southeast by the Dog Bay Line (Williams et al., 1993).

2.1.4. Sampling

Lapointe (1979) and Gales et al. (1989) investigated the palaeomagnetism of the Lawrenceton Formation. Lapointe's sampling was restricted to a local area at the southwest end of the outcrop near Grand Falls–Windsor (Fig. 1). Gales et al. (1989) sampled the lava flows 125 km away at the northeast end of the outcrop on Change Islands. We sampled the Lawrenceton Formation at 13 sites along its outcrop between Grand Falls–Windsor and Port Albert (near Change Islands, Figs. 1 and 2a,b). The sites are distributed around several folds with subhorizontal axes. The orientation of lava flows was deduced from reddened lava flow tops, layering in sediments between lava flows, flattened vesicles and amygdules, and foliations defined by plagioclase phenocrysts. Lava flows varied in orientation from horizontal to slightly over-

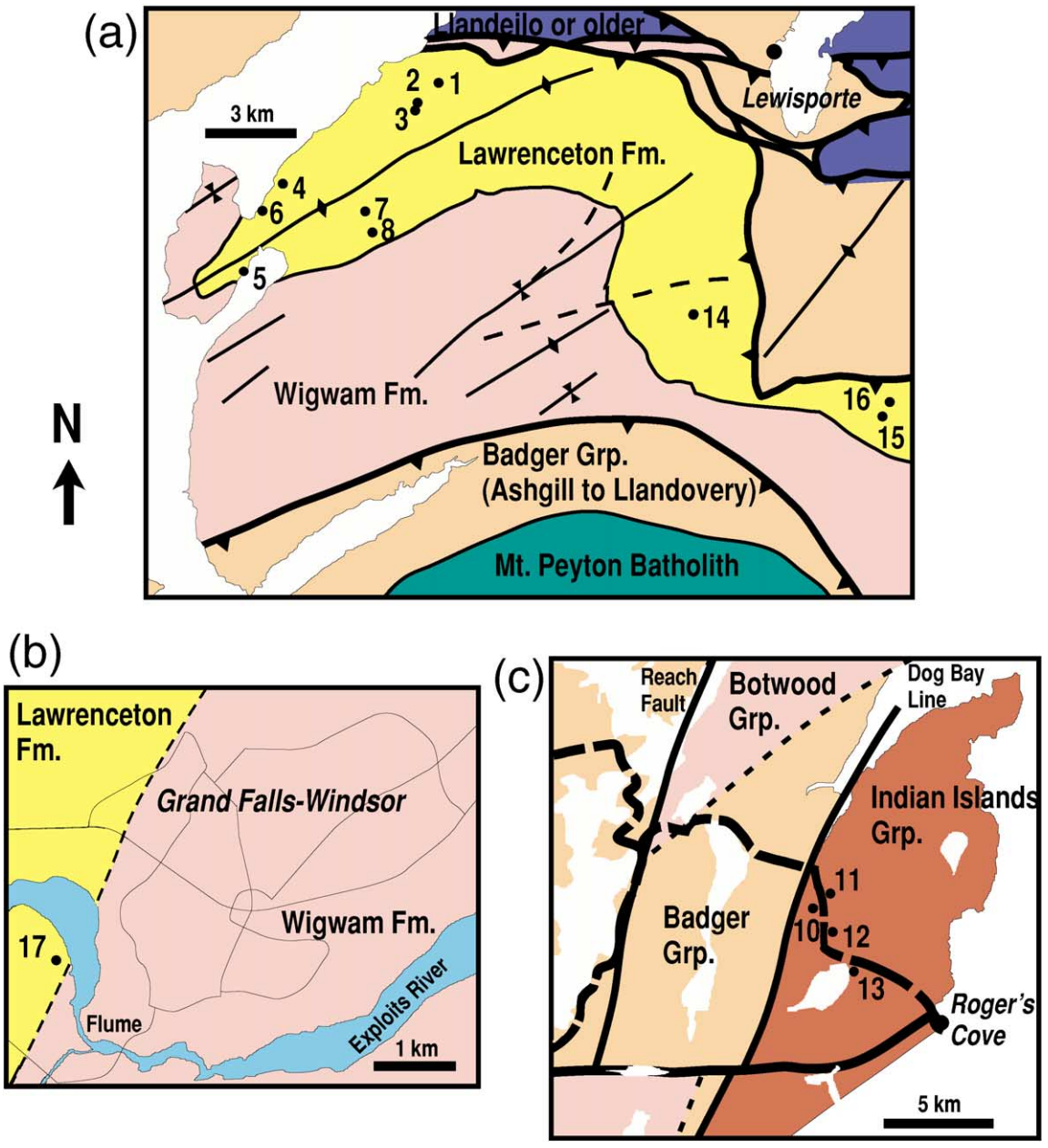


Fig. 2. Simplified geological maps of sampling regions. Sampling sites are numbered. (a) Lewisporte region after Dickson et al. (1995), (b) Grand Falls–Windsor, (c) Horwood Peninsula after Williams et al. (1993).

turned. Between 6 and 15 drill cores, or four to seven block samples, were collected at each sampling site. Drill cores were extracted from the blocks in the laboratory. A total of 178 cylindrical specimens from the Lawrenceton Formation were studied palaeomagnetically.

2.2. Redbeds: Big Indian Pond Formation (Centennial Formation), Indian Islands Group

2.2.1. Rock description

The Silurian Indian Islands Group consists of shallow marine sediments passing upward into subareal redbeds (Williams, 1972, 1993; Williams et al., 1993). The Indian Islands Group lies immediately to the east of the Botwood Group, separated from it by the Dog Bay Line of Williams et al. (1993). Williams et al. described the Dog Bay Line as a Silurian terrane boundary representing the closure of a Silurian seaway. They concluded that considerable dextral movement on the Dog Bay Line led to the juxtaposition of the Indian Islands and Botwood Groups in the late Llandovery or later in the Silurian.

The Big Indian Pond (Centennial) Formation consists of subaerial red sandstones and siltstones of early to mid-Silurian age (Williams et al., 1993; Currie, 1993).

2.2.2. Deformation

Like the Botwood Group, the Indian Islands Group underwent folding in mid- to late Silurian times, followed by brittle thrusting (Williams et al., 1993). The majority of folds trend northeast with subhorizontal axes. The Indian Islands Group is more intensely deformed than the Botwood Group with a strongly developed crenulation cleavage in places.

2.2.3. Sampling

In our sampling of the Big Indian Pond Formation, we selected the least cleaved red sediments. In all, 15 large block samples were collected from four sites. A total of 69 cylindrical specimens were extracted from these in the laboratory and analysed for palaeomagnetism. Although the sampled strata were steeply dipping or even overturned, way up was readily established in the field from ripple marks and associated cross-lamination.

3. Laboratory methods

All palaeomagnetic and rock-magnetic experiments were carried out in the palaeomagnetics laboratory at the Lamont–Doherty Earth Observatory of Columbia University in New York. Measurements of anisotropy of magnetic susceptibility (AMS) were made using a Kappabridge KLY-2 susceptibility meter and 12 different specimen orientations. Progressive thermal demagnetisation experiments were carried out using a custom-built furnace mounted in a magnetically shielded room. The residual field in the furnace did not exceed 10 nT. Between 15 and 25 temperature steps were used up to 690 °C. Progressive alternating field demagnetisation was performed on a small number of rhyolite specimens using a Schonstedt AF demagnetiser. Remanence measurements were made using a 2G Enterprises cryogenic magnetometer mounted in the field-free room. Hysteresis properties of the rocks were determined using a Princeton Instruments Alternating Gradient Force magnetometer 2900 (Micromagnetometer). An Alpha Precision Instruments horizontal Curie balance, a Bartington bulk susceptibility meter and an ACS pulse magnetiser were utilised to determine the magnetic mineralogy of the sampled lithologies. The rock-magnetic experiments were complimented by reflected- and transmitted-light optical microscopy on polished thin sections of representative specimens.

4. Multicomponent remanence carried by the Lawrenceton Formation lava flows

4.1. Progressive demagnetisation

Progressive thermal and alternating field demagnetisation experiments were carried out on 178 specimens from 13 sampling sites distributed through the Lawrenceton Formation (sites 1–9, 14–17). Alternating field demagnetization to peak fields of 100 mT had little effect on the remanence and, therefore, thermal demagnetization was mainly used to isolate individual remanent components. Linear and planar elements in demagnetisation data were identified and their confidence limits defined using the LINEFIND statistical package of Kent et al. (1983).

The lavas carry at least three components of natural remanent magnetisation (NRM, Fig. 3). The component with the lowest stability records the recent earth's field in Newfoundland and is not discussed further. The intermediate ('I': 200–580 °C) and high ('H': 580–685 °C) stability remanence components are both of dual polarity and both are early Silurian. I

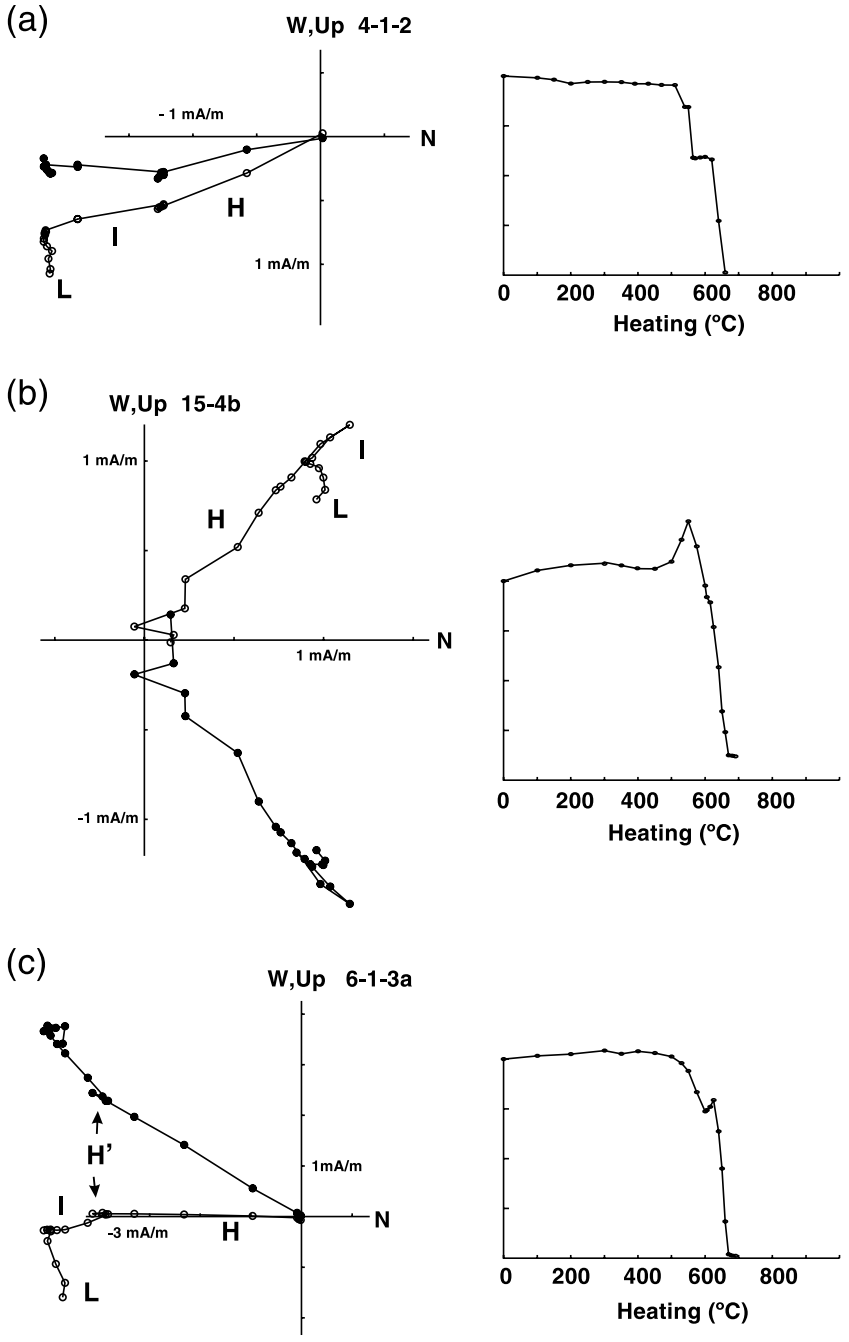


Fig. 3. Zijderveld plots and normalised demagnetisation curves of progressive demagnetisation data from volcanics of the Lawrenceton Formation. Solid (open) symbols denote data projected into the horizontal (vertical) plane. (a) Lava flow containing magnetite and hematite. 'I' and 'H' have the same polarity but are slightly oblique. (b) Like (a), but 'I' and 'H' have opposite polarities and are exactly oppose each other. (c) Like (a), but a component 'H' is removed above the Curie point of magnetite prior to removal of the bulk of 'H'.

and H have unblocking temperature ranges consistent with magnetite and hematite hosts, respectively.

4.2. Optical microscopy

In the Lawrenceton Formation, both magnetite and hematite are in evidence in polished thin section under reflected light. Various morphologies of magnetite are present as discrete, skeletal and well-crystallized grains and as composite grains with ilmenite. Fig. 4a shows an early skeletal magnetite grain adjacent to an ilmenite grain. The magnetite altered to hematite, and most of the ilmenite oxidized to hematite and rutile, though a small amount of ilmenite remains. Fig. 4b shows an early magnetite grain with oxidation–exsolution of ilmenite on the {111} crystallographic planes. Only a small volume of the original magnetite and ilmenite survived later alteration. Most of the trellis ilmenite lamellae were removed by solution, and much, but not all, of the magnetite was replaced by hematite. It is likely that oxidation–exsolution of the magnetite grains producing the ilmenite lamellae took place during initial cooling of the lava, while solution of the ilmenite and oxidation of much of the magnetite to hematite took place later. In some thin sections, magnetite was first altered to maghemite, and then to hematite. The hematite grains that form pseudomorphs of early magnetites are up to 300 μm in diameter. At an early stage in the cooling of the lava flow, abundant hematite also formed on the rims of vesicles in the reddened tops of flows (Fig. 4c) and as rims on phenocrysts (Fig. 4d).

Fine-grained crystalline hematite is present in large quantities in the matrix (Fig. 4e), at least in part due to the breakdown of iron silicates. The time-scale for the formation of hematite from the breakdown of silicates is hard to determine but may have been short in a highly oxidising environment with significant fluid flow. Some of the larger hematite grains may have taken a long time to develop (Fig. 4f), perhaps during low temperature burial metamorphism.

In summary, there is petrological evidence for primary magnetite in small quantities in the form of relic grains, for the early genesis of hematite around vesicles and phenocrysts, and for the later breakdown of magnetite and iron silicates to hematite over an unknown duration.

4.3. Rock magnetism

The presence of magnetite and hematite is confirmed by rock-magnetic experiments. Curie balance results show Curie points at approximately 580 °C (magnetite) and 680 °C (hematite). Isothermal remanent magnetisation (IRM) acquisition curves indicate magnetic phases saturating at 0.2 T (magnetite) and >1 T (hematite).

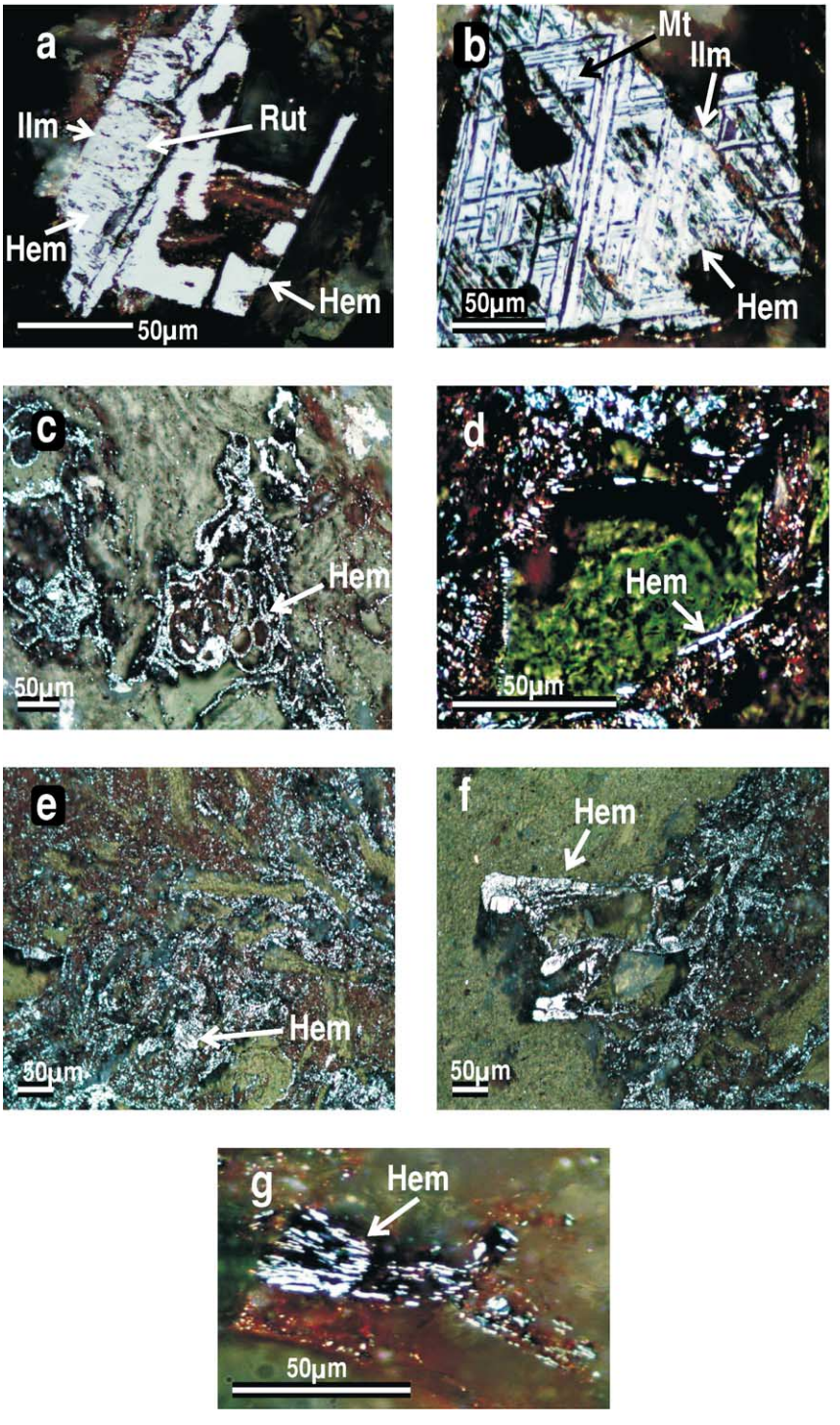
Hysteresis loops for the lavas have wasp-waisted and lozenge shapes (Fig. 5). The wasp-waisted shape, observed in the least oxidised lavas where both magnetite and hematite are observed in thin section (Fig. 4b), results from the large coercivity contrast between magnetite and hematite. The ratio of saturation remanence to saturation magnetisation (M_{rs}/M_s) for these specimens is between 0.17 and 0.39, while the ratio of coercivity of remanence to coercive force (Hcr/Hc) is between 4.60 and 6.70.

The lozenge-shaped hysteresis loop, observed in the strongly oxidised lava flow tops where only hematite was identified in thin section (Fig. 4c and d), indicates a single high-coercivity mineral phase such as hematite. In this case, the ratio of saturation remanence to saturation magnetisation (M_{rs}/M_s) ranges between 0.62 and 0.76, and the ratio of coercivity of remanence to coercive force (H_{cr}/H_c) range between 0.92 and 1.70.

4.4. AMS

All specimens were measured for the anisotropy of magnetic susceptibility (Fig. 6a). Almost all specimens possess a weak flow-parallel magnetic foliation

Fig. 4. Photomicrographs of polished thin sections from the Lawrenceton (a–f) and Big Indian Pond Formations. (a) Hematite (bright white) pseudomorphing skeletal magnetite. Adjacent to hematite, an ilmenite grain has oxidized to hematite + rutile (dark grey). (b) Large magnetite grain, mostly altered to hematite, though internal to the crystal small areas of magnetite (grey) are preserved. Original ilmenite lamellae have been removed by later solutions. (c) Sample from top of lava flow with hematite rimming vesicles. (d) Hematite as rims on phenocrysts, and finely crystalline in matrix. (e) Abundant crystalline hematite rimming silicates. (f) Well-developed hematite at rims of phenocrysts and along edges of matrix silicates. (g) Well-developed crystalline hematite formed from the breakdown of silicates in the Indian Pond Formation. All photomicrographs were made in combined transmitted- and reflected-light using cross polars and a blue filter.


a
Ilm
Rut
Hem
Hem
50µm
b
Mt
Ilm
50µm
Hem
c
Hem
50µm
d
Hem
50µm
e
Hem
30µm
f
Hem
50µm
g
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50µm

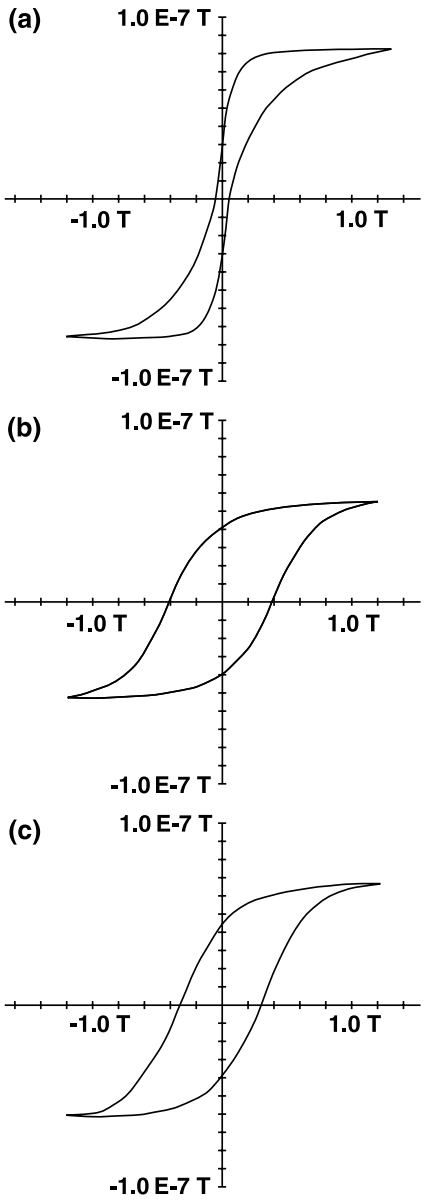


Fig. 5. Hysteresis plots of two lava samples from the Lawrenceton Formation (a, b). The lava samples show a variety of distinctive loops depending on oxidation state. Least oxidized lavas have "wasp-waisted" shapes (a) indicating presence of magnetite and hematite, while more oxidized lavas are dominated by higher coercivity hematite (b). Sedimentary rock samples from the Big Indian Pond Formation are dominated by hematite as shown by "lozenge-shaped" hysteresis loops (c). Horizontal axis is applied field (H) in tesla; vertical axis is magnetization in A m^2/kg.

(P3 (F) < 1.08) consistent with vesicle shapes, phenocryst alignments and other primary textures. When the principal axes of AMS ellipsoids are unfolded to restore them to their original prefolding orientations, palaeo-horizontal, minimum susceptibility axes are vertical and maximum susceptibility axes scatter in the northwest and southeast quadrants (Fig. 6b and c). The alignment of minimum susceptibility axes confirms field measurements of the palaeo-horizontal. The crude alignment of maximum axes may reflect a primary flow-related fabric in the lava flows or a weak tectonic stretching developed during concentric folding.

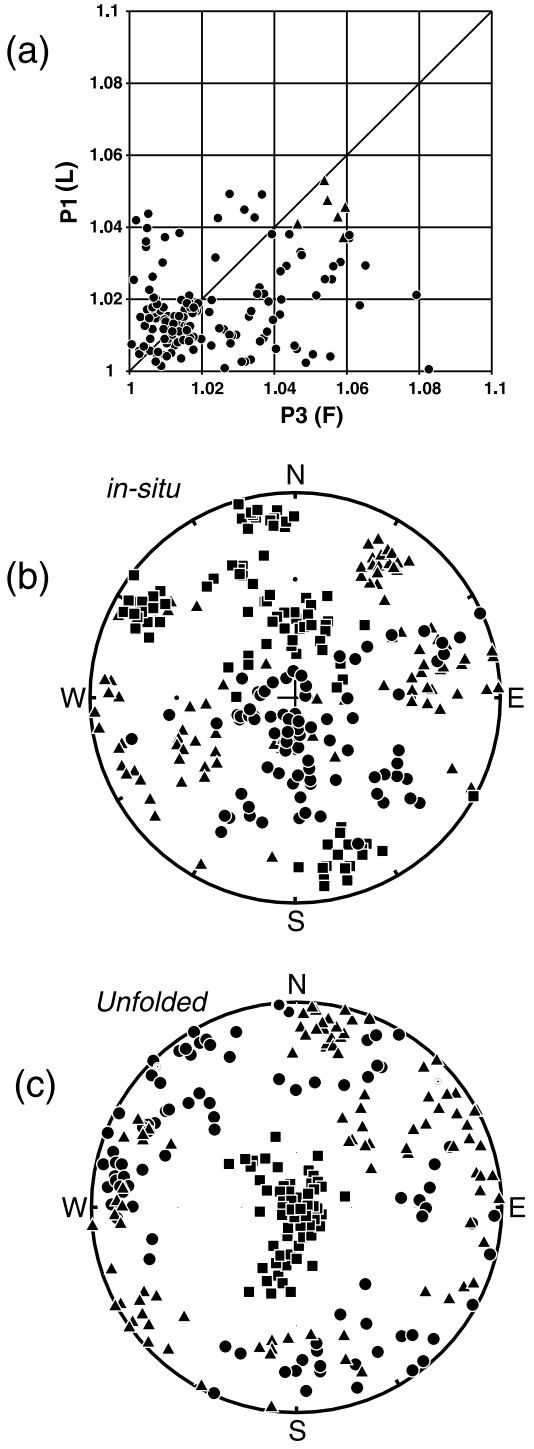
The lava flows at site 9 near the Reach Fault have a strongly developed spaced cleavage with moderate dip to the southeast and a shallow northeast-trending mineral lineation. The magnetic susceptibility ellipsoids are in alignment with tectonic fabrics measured in the field and may relate to strain imposed by transcurrent motions on the Reach Fault. With the exception of site 9, the magnetic fabrics in the lavas are consistent with primary textures commonly observed in undeformed lava flows. It is, therefore, unlikely that remanence directions in the sampled lava flows were altered by penetrative deformation. Because of the well-developed cleavage, the results from site 9 are not used in the determination of overall mean directions or fold tests of palaeomagnetic stability.

4.5. Structural corrections

Folding and differences in remanence polarity from lava flow to lava flow are responsible for the wide scatter in site-level 'I' (magnetite) and 'H' (hematite) remanence directions (Fig. 7a and b; Table 1). Correction for folding translates both sets of directions into northeast up or southwest down directions (Fig. 7c and d, Table 1).

Corrected results from two sites (15 and 16) deviate by 90° in azimuth from the general trend. This discrepancy matches a local swing in strike of the lava flows (Figs. 1 and 2a) and can be explained in terms of local tectonic rotation about a vertical axis. Vertical axis rotations have been observed elsewhere in the Botwood and Springdale Groups (Stamatakos et al., 1995; Potts et al., 1993).

Palaeomagnetic fold tests were applied separately to the 'I' (magnetite) and 'H' (hematite) components



(Table 1, 'fold tests'). Both magnetic components pass the fold test at the 95% confidence level and, therefore, are taken to be older than the middle–late Silurian folding. The observation of relative rotation of 'I' and 'H' between sites 15 and 16, and the rest of the sampled lava flows support the interpretation that both components were acquired prior to deformation.

4.6. Dual polarity

The palaeomagnetic fold tests establish the ages of both remanence components as Silurian, consistent with the observation of both magnetisation polarities in both components. The 'I' and 'H' remanence components have similar directions and for the most part flip polarity in unison from lava flow to lava flow. This implies progressive acquisition of 'I' and 'H' through the sequence of lava flows and a short geological time interval between the formation of the two components in each flow, shorter than the average duration between field reversals. 'I' and 'H' have opposite polarity at sites 5 and 15 (Table 1), indicating that the two remanence components were not acquired synchronously.

4.7. Remanence modes

The reversal patterns for 'I' and 'H' components imply that both were acquired progressively as successive lava flows were emplaced. Remanence component 'I', carried by magnetite, is taken to be a primary thermal remanent magnetisation (TRM) because primary magnetite is seen in thin section. Component 'H' is carried by hematite and is best developed in the reddened lava flow tops where secondary hematite is abundant. 'H' is, therefore, a near-primary chemical remanent magnetisation (CRM) formed while the lava flow top was exposed to steam alteration, weathering, or shortly after burial. The reversal pattern in 'H' rules out protracted CRM

Fig. 6. Anisotropy of magnetic susceptibility of samples from the Lawrenceton Formation. (a) Flinn diagram (χ_{max}/χ_{min} vs. χ_{int}/χ_{min}). Triangles: data from strained rocks at site 9. (b) Equal angle projection of maximum (square), intermediate (triangle) and minimum (circle) principal susceptibility axes. (c) Principal susceptibility axes corrected for folding.

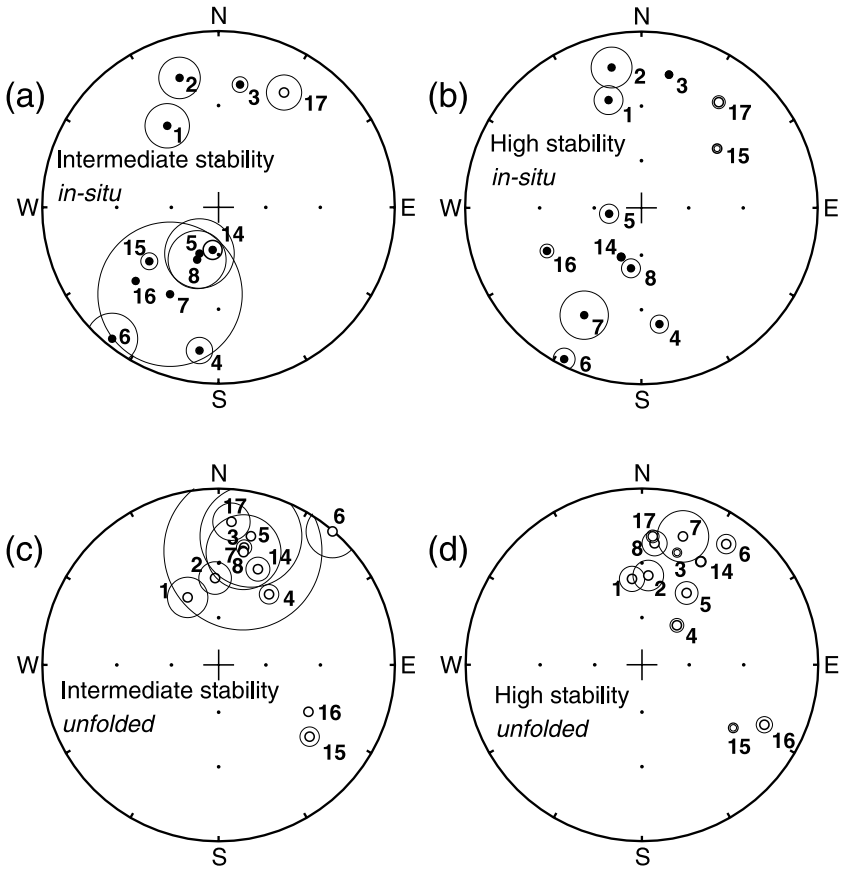


Fig. 7. Site-level remanence component directions for the Lawrenceton formation. (a) 'I' component in situ, (b) 'H' component in situ, (c) 'I' component unfolded, (d) 'H' component unfolded. Equal angle projection.

acquisition. The ages of 'H' and 'I' are interleaved passing up through the lava flow sequence and both 'H' and 'I' can be considered 'primary'. This is supported by positive conglomerate tests on magnetite and hematite components carried by boulders of lava (probably Lawrenceton Formation) in the overlying Wigwam Formation (Buchan and Hodych, 1992).

We did not find a contact between lava flows that could be sampled for a baked contact test and therefore the effect of baking on remanence was not evaluated.

A remanence component with discreet stability range immediately above the Curie point of magnetite was isolated in samples from sites 3, 6 and 7 (labelled 'H′ ' in Fig. 3c). The component occurs in normal and reversed polarity forms and is similar in direction to 'I' and 'H'. It can only be isolated when it opposes both 'I' and 'H' and may, therefore, be widespread in the lava flows. The remanence is apparently carried by hematite and presumably chronicles a minor magnetisation event akin to that of 'H'. Its direction is difficult to determine precisely and it is not considered further.

4.8. Cancellation of secular variation and the early Silurian palaeofield direction

'I' and 'H' constitute punctuated out-of-phase recordings of the palaeomagnetic field during accumulation of the sampled lava flows. The directions of 'I' and 'H' can be distinguished at the site (lava flow) level but overall mean directions are indistinguishable (Table 1, mean directions A and B) implying adequate averaging of secular variation in both data sets. The

Table 1
Palaeomagnetic results from the Lawrenceton Formation, Botwood Group

Site	Lithology	Strike/ dip	χ (SI 10^{-6} (N))	I (200–575 °C): in situ, original polarity					H (>575 °C): in situ, original polarity					I: D/I, unfolded, normal polarity	H: D/I, unfolded, normal polarity
				D/I	N	k	α_{95}	Polarity	D/I	N	k	α_{95}	Polarity		
1	Maroon rhyolite	205/90	356 (21)	328.0/33.3	10	19.8	11.1	N	342.7/24.3	19	26.5	6.6	N	334.6/ – 44.4	353.6/ – 37.9
2	Maroon rhyolite	210/75	511 (7)	343.5/15.5	9	37.6	8.5	N	347.7/11.3	9	43.7	7.9	N	357.1/ – 37.8	004.3/ – 36.3
3	Maroon rhyolite	218/80	296 (8)	010.4/19.8	6	399.4	3.4	N	011.2/14.4	10	1114.4	1.4	N	011.9/ – 21.8	017.6/ – 22.7
4	Maroon rhyolite	225/60	531 (4)	187.2/11.1	4	325.5	5.1	R	171.6/23.1	4	493.5	4.1	R	035.3/ – 38.1	042.0/ – 56.7
5	Grey rhyolite	095/43	378 (13)	201.2/58.1	8	8.2	20.6	R	261.5/68.4	10	65.9	6.0	NR	013.8/ – 16.4	032.2/ – 38.3
6	Grey rhyolite	218/62	388 (16)	218.6/02.4	13	24.2	8.6	R	207.5/02.3	13	126.7	3.7	R	040.4/ – 00.6	035.1/ – 10.3
7	Pink tuff	33/32	208 (5)	208.5/31.0	3	14.6	33.5	R	208.7/21.0	5	53.4	10.6	R	011.7/ – 23.4	017.9/ – 15.3
8	Red tuff	087/32	194 (14)	201.1/54.2	5	21.6	16.9	R	190.9/52.1	14	52.6	5.5	R	012.1/ – 23.9	006.1/ – 20.7
9	Grey lava flow interior. Penetrative deformation	040/45 ?	7543 (6)	[173.1/32.0]	[6]	[420.1]	[3.3]	[R]	[169.7/29.5]	[6]	[232.1]	[4.4]	[R]	[345.5/03.6] ?	[344.1/07.2] ?
14	Brick red lava flow top	128/35	373 (18)	186.6/62.2	23	28.4	5.8	R	203.8/56.8	26	136.5	2.4	R	022.0/ – 29.6	029.9/ – 22.1
15	Grey lava flow interior	265/94	870 (14)	[231.3/37.2]	[16]	[70.5]	[4.4]	[R]	[051.1/ – 32.9]	[21]	[445.3]	[1.5]	[N]	[128.7/ – 23.5]	[124.6/ – 25.4]
16	Grey lava flow interior	265/94	710 (10)	[228.4/25.8]	[1]	[–]	[–]	[R]	[246.2/28.8]	[8]	[282.0]	[3.3]	[R]	[118.1/ – 30.3]	[116.1/ – 14.4]
17a	Brick red lava flow top	020/82	333 (11)	30.2/ – 16.4	12	36.2	7.3	N	36.2/ – 16.8	19	150.9	2.7	N	004.8/ – 11.9	004.8/ – 17.7
17b	Grey lava flow	020/82	2496 (4)												
Mean direction for I in situ:				11.8/ – 17.8	10	4.4	26.2	N							
Mean H in situ:				13.4/ – 20.3	10	4.6	25.5	N							
Mean I and H in situ:				012.6/ – 19.0	20	4.7	16.9	N							
A) Mean I unfolded:				013.4/ – 25.8	10	15.5	12.7	N							
B) Mean H unfolded:				017.7/ – 28.5	10	17.6	11.9	N							
C) Mean I and H unfolded:				015.5/ – 27.2	20	17.6	8.1	N	Site = 49.1 °N/304.3 °E, N pole = 25 °N, 107.7 °E; dp/dm = 4.8 °/8.8 °, Palaeolatitude = 14.4 °S						

Fold tests: I positive fold test (N = 10): $K_{unfolded}/K_{in\ situ}$ = 15.5/4.4 = 3.5 ≫ 2.2 critical ratio for 95% significance. H positive fold test (N = 10): $K_{unfolded}/K_{in\ situ}$ = 17.6/4.6 = 3.8 ≫ 2.2 critical ratio for 95% significance. Combined H and I positive fold test (N = 20): $K_{unfolded}/K_{in\ situ}$ = 17.6/4.7 = 3.8 ≫ 1.7 critical ratio for 95% significance.

χ (N) = mean magnetic susceptibility (number of specimens). I and H = remanence components I and H. D/I: mean remanence declination/inclination, N: number of specimens, k: Fisher's (1953) precision parameter, α_{95}: half-angle of the cone of 95% confidence, Polarity: normal or reversed. Data in square brackets are excluded from overall mean directions for reasons given in the text. The simple fold test of McElhinny (1964) is used because it is straightforward and the convergence of data upon unfolding is obvious.

magnetite ('I') and hematite ('H') components evidently record the same palaeofield and, therefore, can be combined into a single mean direction that best represents the overall primary field direction for the Lawrenceton Formation lava flows (Table 1, mean direction C). This is an unusual situation, because of dual components in single samples, where the number of observations underlying a mean direction (222) exceeds the number of specimens yielding interpretable data (129).

5. Multicomponent remanence in the Big Indian Pond (Centennial) Formation redbeds

5.1. Progressive demagnetisation

Progressive thermal demagnetisation experiments were carried out on 69 specimens from four large sampling sites in the Big Indian Pond Formation redbeds (sites 10–13). Although the data are noisy compared with the lavas, two remanence components were identified. The majority of these were from sites 10 and 11 where samples were taken around fold structures and in both cleaved and uncleaved rocks. The low stability component points north and is a recent viscous remanence (Fig. 8). The stable remanence has a bimodal unblocking temperature spectrum spanning 200–680 °C and is directed steeply down to the northeast and southeast (Fig. 9a and b; Table 2).

5.2. Rock magnetism

The predominant remanence carrier has a Curie point at around 680 °C, did not completely saturate in applied fields up to 1 T and has a high coercivity ($M_{rs}/M_s = 0.7$ and $H_{cr}/H_c = 1.5$, Fig. 5c). We interpret this remanence as being carried by hematite.

5.3. Optical microscopy

Hematite is present in the form of well-crystallised plates as rims around detrital grains and as hematite pigment. The presence of specular hematite in and around relic iron silicate grains indicates in situ production of diagenetic hematite (Fig. 4g). In addition, titanohematite was identified in the form of detrital grains. The hematite and titanohematite account for the bimodal unblocking temperature spectrum in the redbeds.

5.4. Structural correction and anisotropy of magnetic susceptibility

Four block samples were collected around a reclined, open fold in steeply dipping strata at site 10 (blocks 4–7, Table 2). The interlimb angle of the fold is approximately 150° and the fold axis plunges at 30° towards the southwest (Table 2). Cleavage is poorly developed in the fold, but a cleavage–bedding intersection measured at site 11 is subparallel to the directly measured fold axis.

Anisotropy of magnetic susceptibility was measured on all specimens (Fig. 10). As expected, the fine-grained redbeds at site 10 have a weak magnetic foliation subparallel to bedding (P3 (F) = 1.02–1.10). The dominant mineral phase with a significant magnetic susceptibility in the redbeds is hematite and, therefore, the anisotropy of susceptibility is attributed to a primary sedimentary fabric involving hematite grains.

The cleaved redbeds at site 11 have a weakly developed foliation in approximate alignment with bedding (P3 (F) = 1.01–1.05) and a relatively well-defined magnetic lineation (P1 (L) = 1.02–1.04). The magnetic lineation is in alignment with locally measured bedding-cleavage intersections (i.e. intersection lineations) and the fold axis at site 10. This alignment implies a significant tectonic ingredient to the bulk magnetic anisotropy of the cleaved redbeds.

The rocks at sites 12 and 13 are strongly cleaved and this is reflected in the magnetic anisotropy of the sampled material (Fig. 10). The remanence structure in these samples was incoherent.

5.5. Fold test

A subset of palaeomagnetic directions from around a plunging fold at site 10 (blocks 4–7, Fig. 9a) were tilt corrected through the application of a correction for plunge followed by small increments of unfolding (Table 1). The remanence directions converged and diverged again during unfolding (Fig. 9c, squares in Fig. 9d). The tightest grouping, at 55% unfolding [Fisher's (1953) $k = 67$], is significantly tighter than

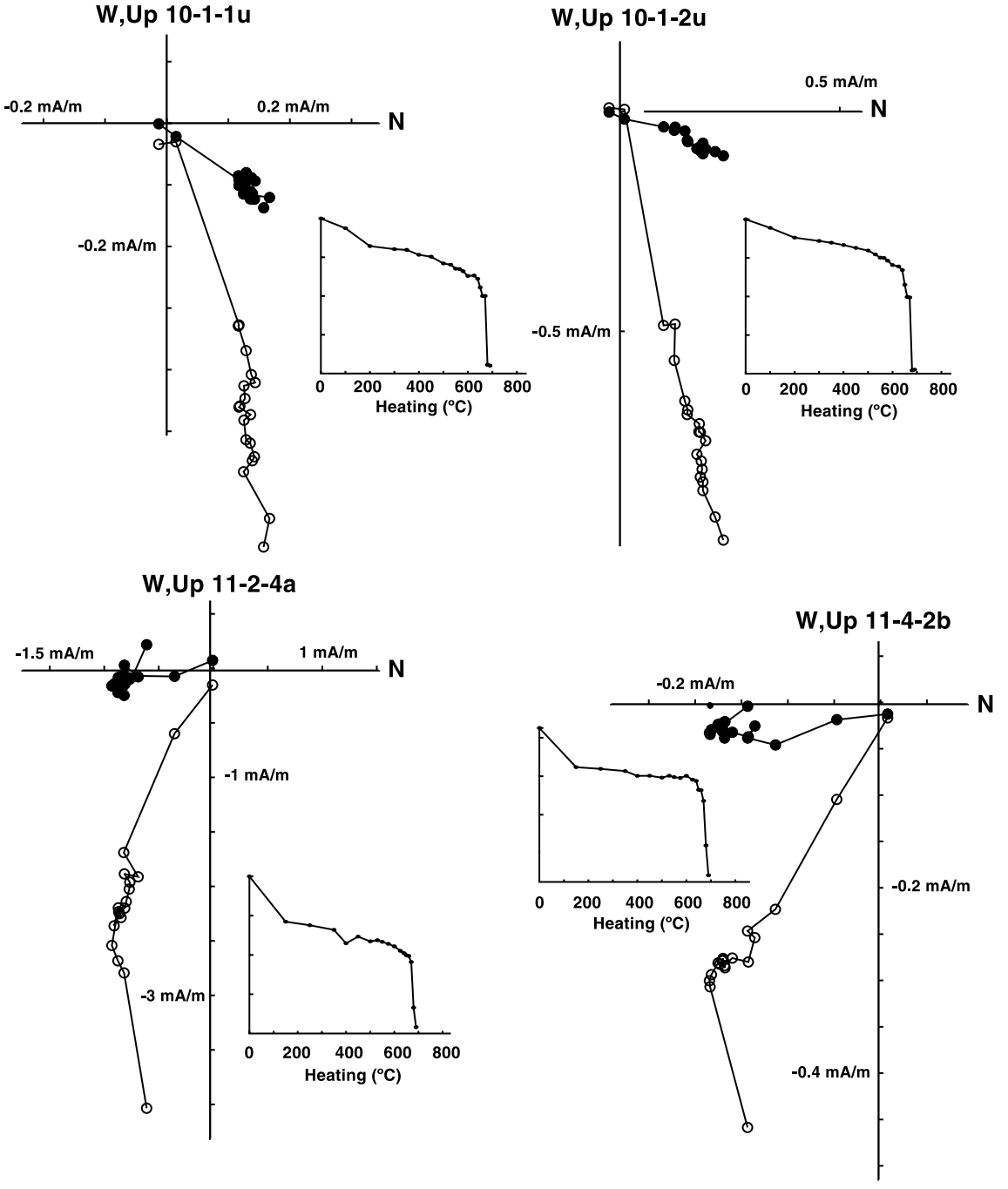


Fig. 8. Zijderveld plots and normalised demagnetisation curves of progressive demagnetisation data from redbeds of the Big Indian Pond Formation.

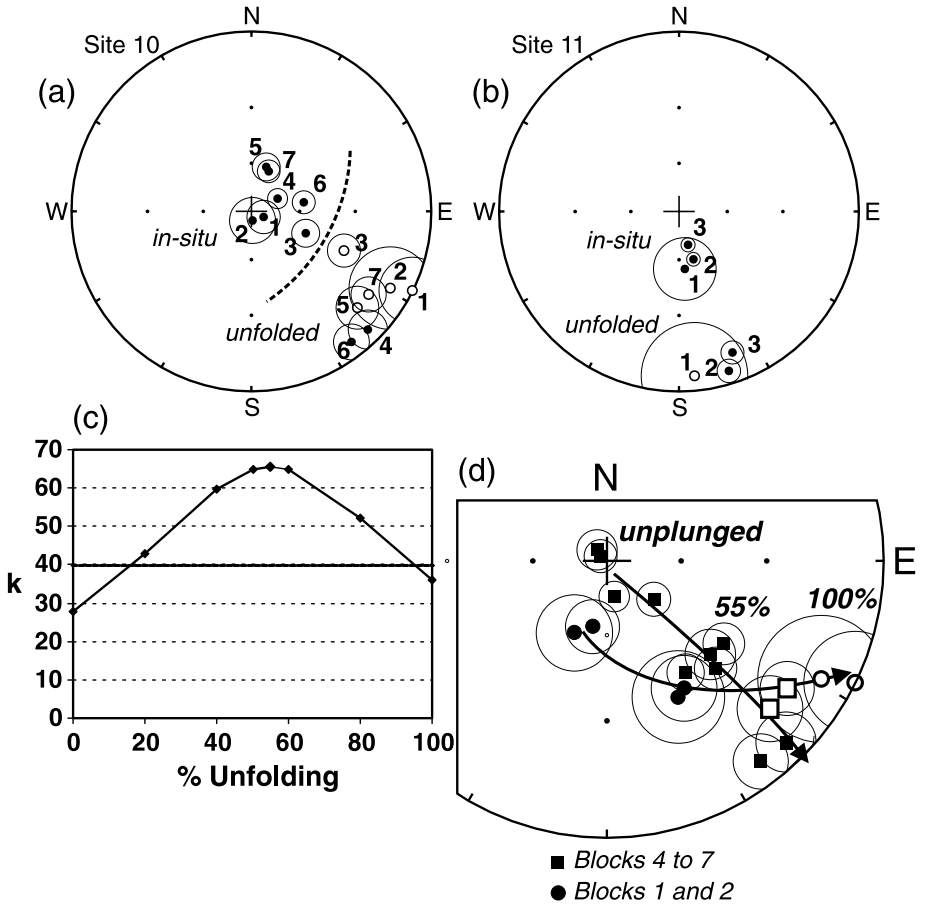


Fig. 9. Block-level remanence directions for the Big Indian Pond Formation. (a) In situ and unfolded directions for site 10. (b) In situ and unfolded directions for site 11. (c) Change in Fisher's (1953) *k* for remanence directions from blocks 4–7, site 10, upon correction for folding ('unfolding'). (d) Change in direction of remanence from blocks 1 and 2 (circles) and 4–7 (squares), site 10, upon correction for folding.

both the folded (in situ) and fully unfolded groupings with significance $p = 0.05$.

A second group of remanence directions from blocks 1 and 2 (circles in Fig. 9d) several metres from the fold were compared with data from the fold. The two groups of directions converge midway under incremental unfolding and confirm an apparent syn-folding origin for the remanence.

The remaining remanence directions for the sediments come from site 11. Incremental unfolding of these directions produces a track separate from but parallel to the unfolding track for site 10. This could be caused by an undetected vertical axis rotation between sites 10 and 11.

5.6. 55% unfolding

The fold test implies a syn-folding origin for the remanence in the sediments. Field observation of axial planar cleavage and anisotropy of susceptibility measurements in the laboratory indicate penetrative deformation of at least some of the sampled sediments. It is well known that penetrative deformation can deflect prefolding remanence directions in the limbs of a fold towards the axial plane so that the remanence appears to be syn-folding when it really is not (e.g. Stamatakos and Kodama, 1991). Even though distinct tectonic fabrics were only detected in part of the sample collection, we cannot rule out the possibility that the

Table 2
Palaeomagnetic results from the Big Indian Pond Formation, Indian Islands Group

Site block	Fold axis (Az./Pl.)	Strike/ dip	χ (SI 10^{-6} (N))	H: in situ, original polarity					H: D/I (unplunged)	H: D/I (55% unfolded)					H: D/I (unfolded)				
				D/I	N	k	α_{95}	Polarity											
10-1	211.6/27.2	031/80	320 (7)	113.6/81.3	6	41.4	10.5	R	192.5/62.7	148.8/33.5					116.1/ − 00.1				
10-2	211.6/27.2	023/90	279 (3)	170.8/84.2	3	73.6	14.5	R	204.4/58.2	152.3/32.0					118.8/ − 07.1				
10-3	211.6/27.2	028/83	377 (5)	[111.7/53.8]	[6]	[70.8]	[8.0]	[R]	[147.2/49.8]	[135.4/06.9]					[112.8/ − 31.5]				
10-4	211.6/27.2	050/80	275 (6)	064.0/71.4	6	118.8	6.2	R	170.2/75.0	144.7/37.3					142.4/05.1				
10-5	211.6/27.2	032/112	359 (8)	018.7/60.7	7	53.5	8.3	R	314.8/83.6	132.1/36.4					132.1/ − 12.9				
10-6	211.6/27.2	060/72	317 (5)	080.1/56.9	5	130.3	6.7	R	130.1/65.6	134.6/32.1					135.3/04.6				
10-7	211.6/27.2	027/106	337 (6)	023.7/62.6	5	129.8	6.8	R	301.2/86.4	125.3/35.1					125.2/ − 12.7				
11-1	211.6/27.2	049/65	393 (5)	173.9/54.5	5	19.3	17.9	R	187.1/30.9	175.8/12.4					174.5/ − 05.0				
11-2	211.6/27.2	043/58	352 (8)	162.8/59.0	8	191.3	4.0	R	182.4/37.4	166.9/21.1					162.5/04.3				
11-3	211.6/27.2	042/61	362 (5)	164.2/68.3	5	316.7	4.3	R	188.8/45.4	166.1/28.9					159.1/10.1				
12-1	–	052/55	346 (3)	[080.9/58.6]	[3]	[1.5]	[180]	[?]	–	–					[113.6/16.4]				
13-1	–	205/53	416 (4)	[310.6/ − 28.3]	[10]	[3.8]	[28.7]	[?]	–	–					[355.8/ − 74.2]				
13-2	–		433 (4)																
Mean site 10 (10-1, 2, 4, 5, 6, 7)				054.0/73.6 Site = 49.1°N, 304.3°E; S pole = 57.8°N, 354.8°E; dp/dm = 24.0°/26.8°	6	21.2	14.9	R	183.8/76.6	139.7/34.8 Site = 49.1°N, 304.3°E; S pole = 12.9°S, 343.1°E; dp/dm = 5.0°/8.6°; Palaeolatitude = 19.2°S	6	81.4	7.5	R	128.3/ − 03.9 Site = 49.1°N, 304.3°E; S pole = 25.6°S, 004.7°E; dp/dm = 5.5°/10.9°	6	39.0	10.9	R
Mean site 11 (11-1, 2, 3)				167.5/60.7 Site = 49.1°N, 304.3°E; S pole = 1.5°N, 313.6°E; dp/dm = 13.6°/17.9°	3	111.2	11.7	R	186.0/37.9	169.8/20.9 Site = 49.1°N, 304.3°E; S pole = 29.4°S, 315.8°E; dp/dm = 8.2°/15.6°; Palaeolatitude = 10.8°S	3	70.1	14.8	R	165.4/03.2 Site = 49.1°N, 304.3°E; S pole: 37.0°S, 322.9°E; dp/dm = 8.5°/17.0°	3	53.5	17.0	R
Incremental fold test (10-4, 5, 6, 7)				201.6/59.1	23	27.9	5.8	R	154.5/83.9	134.4/35.6	23	65.4	3.8	R	134.0/ − 4.3	23	35.9	5.1	R
	0%	20%	40%	50%	55%	60%	80%	100%											
k	27.9	42.8	59.7	64.8	65.4	64.7	52.2	35.9											

55% vs. in situ (N = 27): $K_{55\%}/K_{\text{in situ}}$ = 65.4/27.9 = 2.3 ≫ 1.7 critical ratio for 95% significance. 55% vs. unfolded (N = 27): $K_{55\%}/K_{\text{unfolded}}$ = 65.4/35.9 = 1.8>1.7 critical ratio for 95% significance.

Abbreviations are as in Table 1.

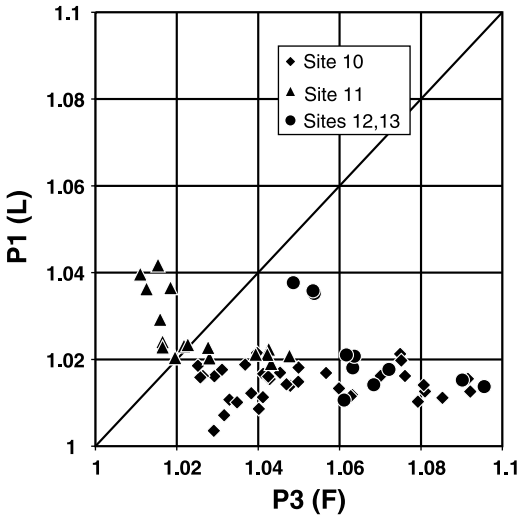


Fig. 10. Flinn diagram of the anisotropy of susceptibility of samples from the Big Indian Pond Formation.

penetrative deformation has altered remanence directions in the sediments. Furthermore, if the remanence is syn-folding, we do not know the attitude of the fold with respect to the horizontal plane midway though its development.

6. Discussion

In Fig. 11, we compare the new data in the form of palaeomagnetic south poles with previously published data from the Central Mobile Belt of Newfoundland and Laurentia. As reference data for Laurentia, we use a subset of the data chosen for palaeomagnetic reliability by Torsvik et al. (1996). The Silurian part of the data set is dominated by results from igneous rocks in Scotland. The only reliable data from North America in this period are from the Wabash Reef limestone in Indiana, USA. New and previously published palaeomagnetic results from the Silurian rocks of the Central Mobile belt define a broad swath cutting the Silurian part of the apparent polar wander (APW) path for Laurentia. The swath is a small circle centred on Newfoundland and testifies to post-early Silurian vertical axis rotation of the studied rock units. The poles lie near the Silurian part of the APW path or to the west of it, indicating a range of orientations from no net rotation to large clockwise rotation of rock bodies depending on location in the Central Mobile Belt. In this study, we are concerned with field inclination and palaeolatitude so we transformed the palaeomagnetic poles in Fig. 11 into predicted palaeolatitudes for the Central Mobile Belt and plotted them against geological time in Fig. 12.

The curve in Fig. 12a depicts the palaeolatitudinal drift of the Newfoundland margin of Laurentia. Published Silurian data from the Central Mobile Belt fall in two groups: those from sediments indicate shallow palaeolatitudes, and those from volcanics indicate higher (southerly) palaeolatitudes. The first point to be noted is that the sediments, and not the volcanics (e.g. Stamatakos et al., 1995), yield palaeolatitudes consistent with a position on the Newfoundland margin of Laurentia. This is because we use an updated path for Laurentia that is slightly different in form to the path used by Stamatakos et al. (1995) and others. The relevant part of our path (430–420 Ma) is defined by results from Scottish granites and intrusive complexes and the Wabash Reef limestone. The series of mean poles (Van der Voo, 1993) used by Stamatakos et al. (1995) define a smoother path than ours because the mean poles average different and fewer results over wide time windows. We regard the newer path in Fig. 11 as the best record of the movement of Laurentia. This means that the results from the majority of the sediments are consistent with it (poles SW, WF79, WF92, SG, SGR) and that the majority of the results from the volcanics are not (poles LF79, LF89).

Our new data for the Lawrenceton Formation lava flows (Fig. 12b, overall result LF, site 15, site 16) are similar to the published results from the sediments and overlap with the reference palaeolatitude curve for the Newfoundland margin of Laurentia. There is no significant discrepancy between these new results from volcanic rocks, and previously published results from sediments (except KG4), and both are consistent with a position of the Central Mobile Belt against the margin of Laurentia.

The new results from the Lawrenceton Formation disagree with previously published data from the same rock formation and from the correlative Springdale volcanics because like the published studies for the Lawrenceton Formation (LF79 of Lapointe, 1979, and LF89 of Gales et al., 1989), remanence directions with intermediate inclinations were also

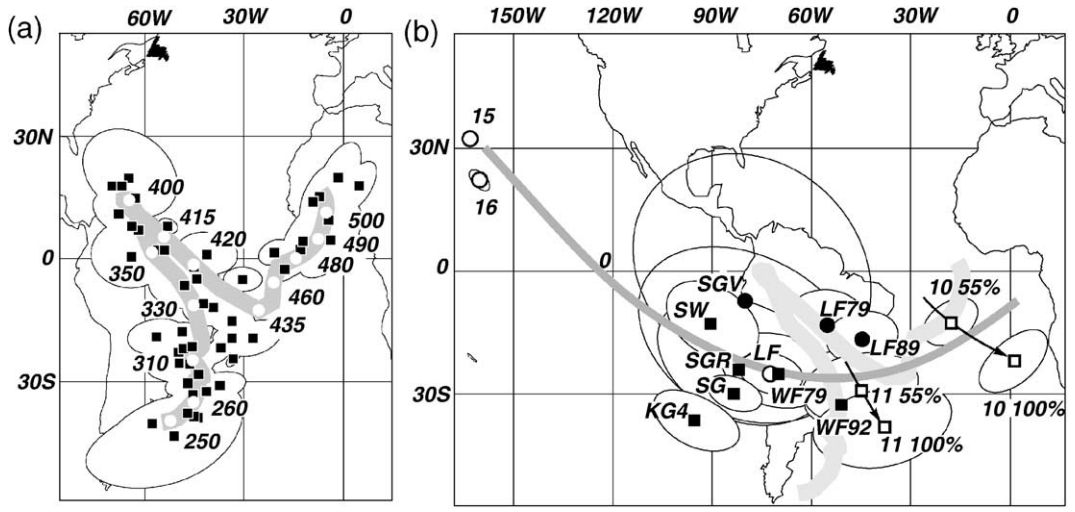


Fig. 11. (a) Reference palaeomagnetic south poles for Laurentia after MacNiocaill and Smethurst (1994) as modified by Torsvik et al. (1996). The 'spherical spline' algorithm of Jupp and Kent (1987) as implemented in the GMAP program of Torsvik and Smethurst (1999) was used to fit a smooth apparent polar wander (APW) path to the data. Ages for the path (My) are indicated. (b) Palaeomagnetic south poles for Silurian rocks of the Central Mobile Belt of Newfoundland superimposed on the APW path for Laurentia. Squares denote results from sediments and circles denote results from volcanics. Palaeomagnetic poles: LF79, WF79: Lawrenceton volcanics, Wigwam sediments of Lapointe (1979); LF89: Lawrenceton volcanics of Gales et al. (1989); KG4: King George IV Lake sediments of Buchan and Hodych (1989); WF92: Wigwam sediments of Buchan and Hodych (1992); SG: Springdale sediments of Hodych and Buchan (1994a); SGV, SGR: Springdale volcanics and sediments of Potts et al. (1993); SW: Springdale and Wigwam sediments of Stamatakos et al. (1995); LF: Lawrenceton volcanics of this study ('15' and '16' are results for sites 15 and 16); '10 55%', '11 55%', '10 100%', '11 100%': partially and completely unfolded results from the Big Indian Pond Formation sites 10 and 11 of this study. The dark grey arc is a small circle centred on Newfoundland and represents the possible positions of early Silurian poles after vertical axis rotation.

identified in the new study. Nevertheless, these remanence directions can be shown to belong to an underlying population of directions with a shallower overall mean direction. Lapointe's (1979) result is based on a reversed polarity hematite remanence from three closely spaced sampling sites near Grand Falls–Windsor. Due to limited sampling, it is unlikely that secular variation and inaccuracy in determination of the palaeo-horizontal have been adequately averaged out in Lapointe's mean direction. The same argumentation applies, but to a lesser degree, to the results of Gales et al. (1989). Reversed polarity hematite remanence directions were obtained from six sites near the top of the Lawrenceton Formation in a tectonically uniform region on Change Islands. Both previous studies reported only reversed polarity remanence in hematite, while we can demonstrate dual polarity remanence carried by hematite and magnetite. This observation reinforces our argument that the earlier studies constitute a biased sample of the true palaeomagnetic record in the Lawrenceton Formation.

The difference between the new result and the steeper result from the correlative Springdale volcanics (SGV, Potts et al., 1993) is insignificant because the Springdale result has a large angular error margin of 28.3° (inclination error limits: +27°/−17°). The Springdale volcanics result is based on data from only five sites with a large between-site scatter.

Our new results from the Big Indian Pond Formation are also compared with other data in Figs. 11 and 12. There is a chance that the palaeomagnetic directions have been altered by penetrative deformation and we can only make an assumption regarding the orientation of the fold midway though its development. Nevertheless, we show in Fig. 11b that 55% unfolding moves the data straight into the arc of Silurian data and further unfolding translates them out again. Therefore, the partially corrected Indian

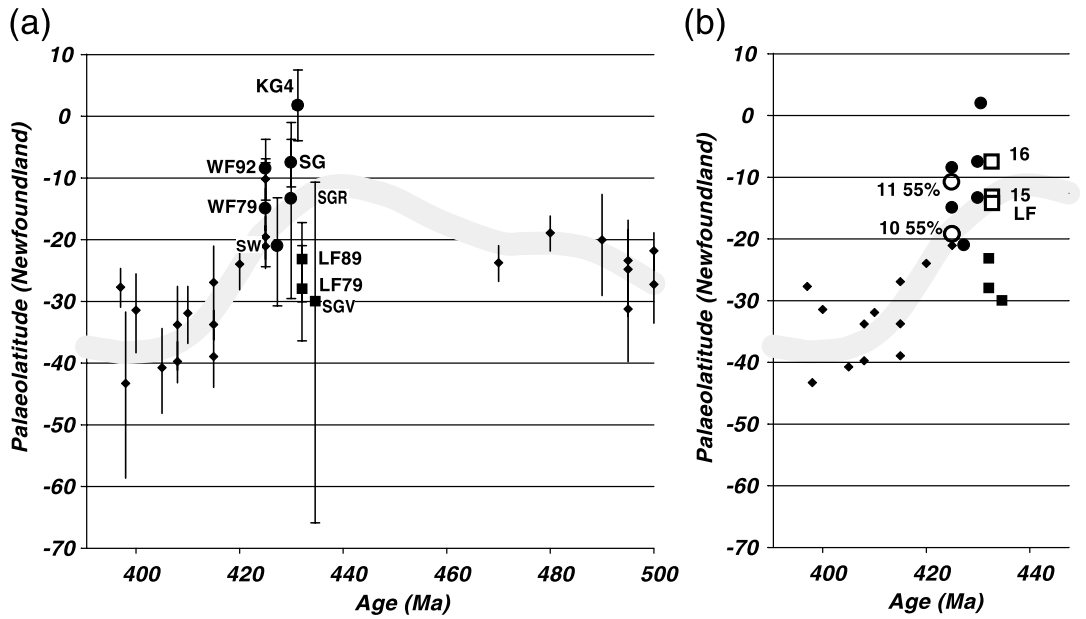


Fig. 12. (a) Change in palaeolatitude of the Newfoundland margin of Laurentia predicted by the reference palaeomagnetic poles of Fig. 11. Reference palaeolatitudes for Laurentia are indicated by diamonds with 95% error bars. Palaeolatitudes predicted by published data from the Central Mobile Belt of Newfoundland are indicated by squares (volcanics) and circles (sediments). Labels for the published data are as in Fig. 11. (b) Palaeolatitudes predicted by the new data of this study (open symbols) compared with the published data (closed symbols).

Pond Formation results (10–55% and 11–55% in Fig. 12b) are consistent with results from other sediments, and the new results from the volcanics.

7. Conclusions

Palaeomagnetic investigation of early Silurian rhyolitic lava flows of the Lawrenceton Formation (Botwood Group) in the Central Mobile Belt of Newfoundland yielded a mean palaeomagnetic direction with declination 015.5°, inclination −27.2° and angular error 8.1° that corresponds to a palaeomagnetic north pole position at latitude 25°N and longitude 107.7°E encompassed by an error oval with half-angles 4.8° and 8.8° (site position 49.1°N, 304.3°E). The new result implies a palaeolatitude for the Botwood Group of 14.4°S in the early Silurian consistent with a position on the Newfoundland margin of palaeo-equatorial Laurentia.

Early-mid Silurian redbeds of the Big Indian Pond (Centennial) Formation of the Indian Islands Group were also studied for palaeomagnetism. Although the redbeds carry an ancient remanent magnetisation, it is complex because of Silurian deformation and cleavage development. The remanence, at first appearances appears to be syn-folding and, therefore, Silurian in age. Adding complexity, penetrative deformation, cleavage development and resulting remanence reorientation might give the illusion of syn-folding remanence when in fact the remanence predates folding. In either case, significant uncertainty surrounds the reconstructed Silurian palaeofield direction for these samples. Nevertheless, it is noteworthy that both reconstructed forms are consistent with the palaeomagnetic result from the Lawrenceton Formation, placing the Central Mobile Belt of Newfoundland close to the Laurentian margin in the Silurian.

The new palaeomagnetic result for the Lawrenceton Formation volcanics supersedes previously published results for the same rock unit that implied steeper palaeofield inclinations. The new result for the volcanics is consistent with all but one of the published results for Silurian sedimentary rocks and, therefore, the long-lived discrepancy between palaeomagnetic results from the two rock types appears to

be solved. One steep palaeomagnetic result remains; that for the Springdale Volcanics. This result has a large angular error envelope that overlaps with the shallower data and therefore it does not challenge our argument for cross-lithology palaeomagnetic conformity.

We observe palaeomagnetic inclination conformity across the Central Mobile Belt in the Silurian, from the Indian Islands Group and Botwood Group in the Northeast to the Springdale Group in the Northwest. This inclination is similar to the field inclination for the Newfoundland margin of Laurentia implied by reference data and, therefore, the palaeomagnetic data unite with strong geological evidence for contraction of this part of the Iapetus Ocean to at most a very narrow seaway by the end of the Silurian (Williams et al., 1993; Robinson et al., 1998).

Although the majority of Silurian data from the Central Mobile Belt of Newfoundland agree in field inclination, the data are spread in azimuth over 150° of arc. Most of the data are displaced in a clockwise sense from the Laurentia reference, implying net clockwise vertical axis rotations of the studied regions relative to Laurentia. That the studied regions are rotated by different amounts indicates relative rotation within the Central Mobile Belt. Major orogen-parallel faults with curved traces and low angle brittle faults were active in the Central Mobile Belt after the magnetisations were acquired in the Silurian rocks. We speculate that the clockwise rotations resulted from displacements along known dextral orogen-parallel faults, or with a promontory of some kind on the margin of Laurentia. Closer examination of the rotations is necessary to establish their structural geology.

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